MILITARY SPECIFICATION

ENCODER, SHAFT-ANGLE-POSITION, CYCLIC-DECIMAL-CODED (FOR TORPEDO TEST SET MARK 281 MOD 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers a contact-type, 19-bit and complement, 36,000-count cyclic decimal-coded, shaft-angle position encoder for use with the Torpedo Test Set Mark 281 Mod 1. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

FSC 7440

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MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals

DRAWINGS

Naval Ordnance Systems Command

2639621	Encoder, Shaft Position
2639726	Indicator, Digital
2639766	Connector, Digital Indicator
2639792	Decoder P. C. Card Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First articles. Unless otherwise specified in the contract or purchase order, three first articles of the encoder shall be manufactured using the methods and procedures proposed for production. The first articles will be tested as specified in section 4 to determine if the proposed production method will yield an item which meets the technical and design requirements of this specification and the contract or purchase order.

3.2 Materials.

3.2.1 Fungus proof. Materials shall be selected in accordance with MIL-STD-454, requirement 4.

3.2.2 Corrosion resistance. Materials shall be corrosion-resistant or suitably treated to resist corrosion.

3.2.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact.

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3.2.4 Nonmagnetic materials. Nonmagnetic materials shall be used where practical.

3.2.5 Fumes. Materials which, under the conditons likely to be encountered in storage or service, liberate deleterious fumes shall not be used.

3.3 Design and construction. The design and construction shall be in accordance with the requirements specified herein and on NAVORD Drawing 2639621.

3.4 Weight. The encoder shall not weigh more than 20 ounces.

3.5 Life. The encoder shall be designed and constructed to withstand a service life of 10,000,000 revolutions of the input shaft or 10 years, whichever occurs first.

3.6 Performance.

3.6.1 Operation. The encoder shall operate properly in the clockwise and counterclockwise directions when connected and monitored in accordance with figure 1.

3.6.2 Count. The encoder shall count continuously and consecutively from 00000 to 35999 increasing when the input shaft rotates counterclockwise, 1000 counts per revolution. For clockwise rotation of the input shaft, the encoder shall count from 35999 to 00000, at 1000 counts per revolution. A "count" is defined as the numerals displayed when the encoder is connected as in figure 1.

3.6.3 Resolution. The resolution of the encoder shall be 1 count or a 0.36° rotation of the input shaft.

3.6.4 Stops. The encoder shall have no stops which prevent continuous rotation of the input shaft in either direction, and the entire encoder count shall repeat every 36 rotations.

3.6.5 Slew speed. The encoder shall not sustain damage nor fail to meet the test requirements of this specification after being subjected to input shaft rotation rates in either direction of 200 revolutions per minute with power applied.

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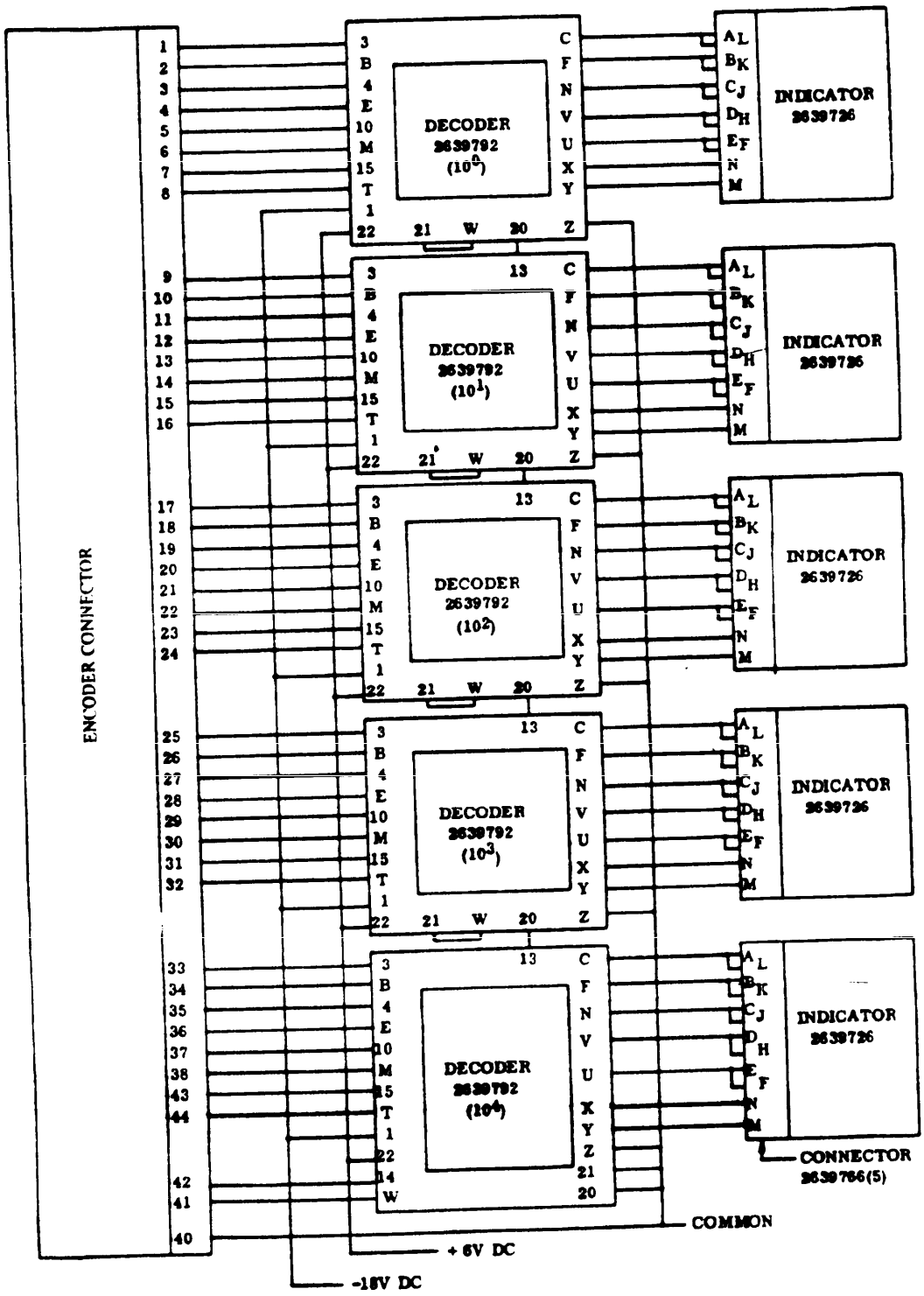


FIGURE 1. APPLICATION SCHEMATIC

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3.6.6 Accuracy. The encoder shall be an absolute position encoder, within ±1 count of the true mechanical position of the input shaft. For example, a rotation of 90° would cause a count change of 250 ± 1 count.

3.6.7 Contact rating. The individual brush-to-disc contact ratings shall be as specified below.

3.6.7.1 Commutating power. The commutating power capability of the encoder brush-to-disc contact shall be a minimum of 15 milliwatts throughout the service life of 3.5. Commutating power is the product of the open-circuit voltage and the contact-brush junction current.

3.6.7.2 Voltage. Each encoder brush shall have a voltage rating of 30 volts across an open contact when the disc is not rotating.

3.6.7.3 Current. Each encoder brush shall have a current rating of 1 milliampere through each closed contact when the disc is rotating.

3.6.8 Insulation resistance. The insulation resistance shall be a minimum of 10 megohms as the input shaft is rotated through 72 revolutions at approximately 5 to 10 revolutions per minute, with 500 volts alternating current at 60 hertz peak-to-peak or 200 volts direct current applied between the case and any or all leads.

3.6.9 Application requirements. The encoder, when properly connected in accordance with figure 1, shall supply the code required by means of closed and open contacts. The commercial equivalent of the required code is Datex Cyclic.

3.6.9.1 Closed contacts. With the encoder connected as in figure 1, the voltage drop measured from the encoder output pins to direct-current power ground on the decoder (NAVORD Drawing 2639792) shall be a maximum of 2 volts when nonrotating and with contacts closed. The voltage across closed contacts shall not exceed 4 volts at any shaft rotation speed up to 50 revolutions per minute.

3.6.9.2 Open contacts. The resistance of open contacts shall be a minimum of 100,000 ohms measured from encoder output pins (negative polarity) to the appropriate common (positive polarity).

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3.7 Code. The encoder output code shall be an unambiguous monostrophic code that requires the mechanical change of only one brush state between any two consecutive numbers. (See 6.3.)

3.8 Physical. The physical characteristics shall be in accordance with NAVORD Drawing 2639621 and 3.8.1 through 3.8.8.

3.8.1 Moment of inertia. The moment of inertia of the input shaft shall not exceed 600 gram centimeters-squared.

3.8.2 Torque. Starting (breakaway) torque shall not exceed 2.0 ounce-inches in either direction.

3.8.3 Radial play. Radial play of the shaft shall not exceed 0.0015 inch (total indicator reading) as measured 0.125 inch (maximum) from the shaft end, with a 4.0-ounce (minimum) reversing load applied 0.5 inch (minimum) from the case.

3.8.4 Runout. Shaft runout shall not exceed 0.001 inch (total indicator reading) as measured within 0.0625 inch of the shaft end, with the case stationary and the shaft rotated through 360° (minimum).

3.8.5 Perpendicularity. The shaft shall be perpendicular to the case within 0.003 inch (total indicator reading) as measured on the 3.375-inch-diameter mounting face with the shaft held stationary and the case rotated through 360° (minimum). The allowable error for this requirement is exclusive of the tolerance effect of 3.8.4.

3.8.6 Concentricity. The shaft shall be concentric to the case within 0.001 inch (total indicator reading) as measured on the 3.375-inch mounting diameter with the shaft held stationary and the case rotated through 360° (minimum).

3.8.7 End play. The shaft end play shall be 0.0004 to 0.002 inch under a 4-ounce reversing load.

3.8.8 Output connections. The output connector shall be in accordance with NAVORD Drawing 2639621. Internal connections to the output connector shall conform to the functional schematic of figure 2.

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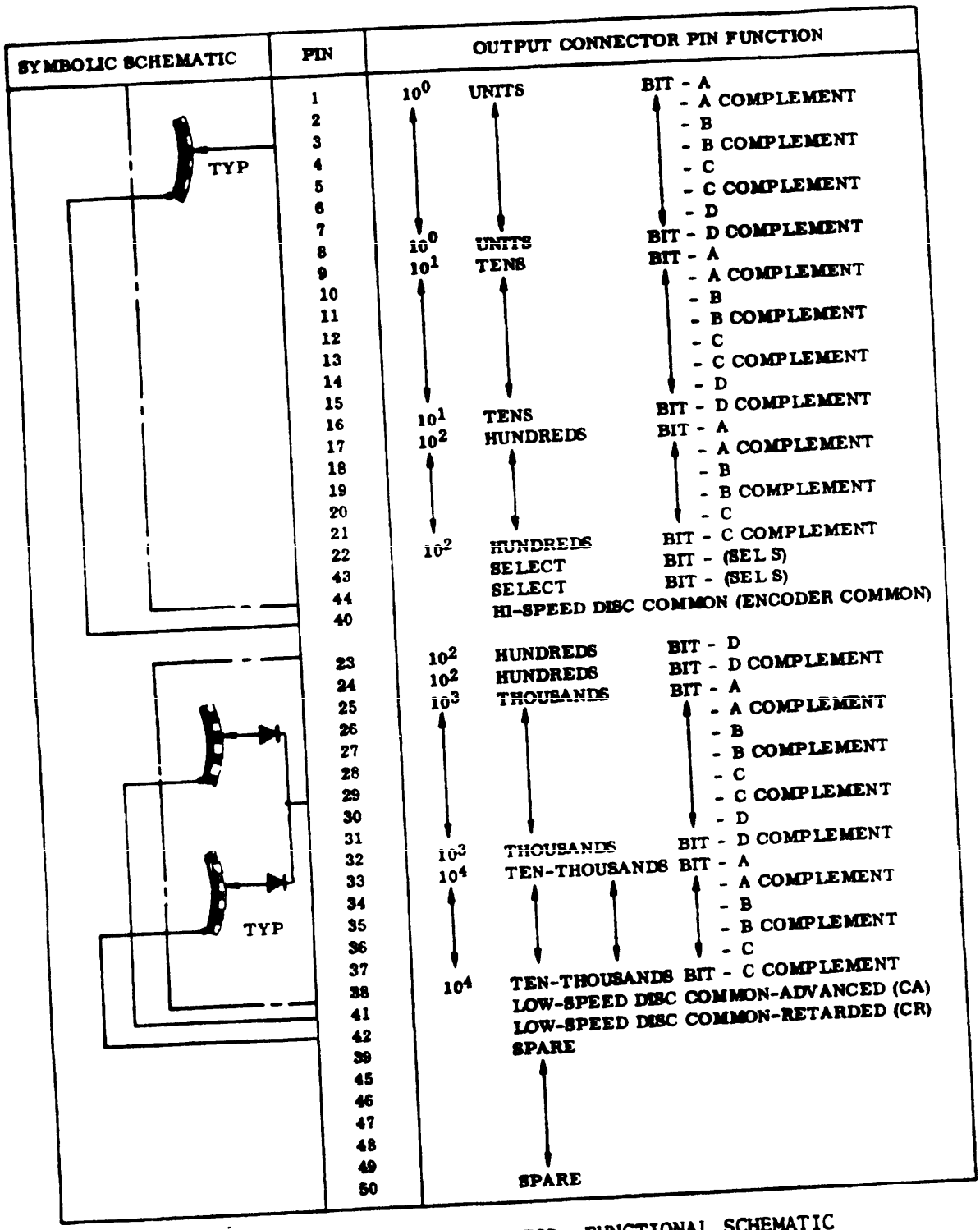


FIGURE 2. OUTPUT CONNECTOR, FUNCTIONAL SCHEMATIC

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3.9 Environmental requirements. The encoder shall meet the other applicable requirements of this specification during or after the tests defined or specified in section 4.

(a) High temperature, operating: 73° Celsius
(b) Low temperature, operating: -15° Celsius
(c) High temperature, storage: 85° Celsius
(d) Low temperature, storage: -62° Celsius
(e) Vibration, operating: MIL-STD-167, type I equipment
(f) Vibration, nonoperating: 2.5 g at 7 to 500 hertz
(g) Shock, nonoperating: 40 g for 18 milliseconds
(h) Humidity, nonoperating: 95 percent relative at 56° Celsius
(i) Humidity, operating: 85 percent relative at 56° Celsius
(j) Salt fog atmosphere, nonoperating.

3.10 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the encoder shall be classified as follows:

(a) First article inspection (see 4.4)
(b) Quality conformance inspection (see 4.5).

4.2.1 First article inspection. Unless otherwise specified in the contract or purchase order, three encoders shall be manufactured using the methods and procedures proposed for production. These encoders shall be subjected to the first article inspection procedure of 4.4.

4.2.2 Quality conformance inspection. The procedure of 4.5 shall be performed on each encoder submitted for acceptance under contract

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(100 percent inspection) to assure that production units meet the requirements of this specification and the contract or purchase order.

4.3 Test conditions. Unless otherwise specified herein, the encoder shall be tested under the following ambient conditions:

(a) Temperature: 25° ± 10° Celsius
(b) Humidity: not exceeding 95 percent relative.

4.3.1 When the equipment is referred to as properly connected, it shall be connected, as shown in figure 1, to decoders (NAVORD Drawing 2639792) and digital indicators (NAVORD Drawing 2639726) and shall have power supplied as follows:

(a) Plus 6.0 ± 0.5 volts direct current
(b) Minus 18.0 ± 0.5 volts direct current
(c) Common for both sources.

4.4 First article inspection procedure.

4.4.1 First article inspection shall be performed on three first articles in accordance with the test schedule shown in table I.

4.4.2 A failure of any encoder in any test or inspection shall be cause for denial of production approval.

4.5 Quality conformance inspection procedure.

4.5.1 Quality conformance inspection shall be performed on each encoder submitted for acceptance under the contract according to the test schedule shown in table I.

4.5.2 A failure of any encoder in any test or inspection shall be cause for rejection of the encoder.

4.6 Visual and mechanical examination. The encoder shall be examined to verify that the design, construction, physical dimensions, and workmanship are in accordance with the applicable requirements of 3.4, 3.8, and 3.10 and NAVORD Drawing 2639621.

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Table I

TEST SCHEDULE

Test	Requirement	Method	First article unit			Quality conformance
			A	B	C	
Visual and mechanical examination	3.4, 3.8, 3.10	4.6	X	X	X	X
Operation	3.6	4.7.1	X	X	X	X
Accuracy	3.6	4.7.2	X	X	X	X
Contact parameter measurement	3.6.9	4.7.3	X	X	X	
Insulation resistance measurement	3.6.8	4.7.4	X	X	X	
Slew speed performance	3.6.5	4.7.5	X	X	X	
High temperature, operating	3.9	4.7.6.1	X	X		
Low temperature, operating	3.9	4.7.6.2	X	X		
High temperature, storage	3.9	4.7.6.3	X		X	
Low temperature, storage	3.9	4.7.6.4	X		X	
Vibration, operating	3.9	4.7.6.5	X		X	
Vibration, nonoperating	3.9	4.7.6.6	X		X	
Shock	3.9	4.7.6.7	X	X		
Humidity	3.9	4.7.6.8	X	X		
Salt fog	3.9	4.7.6.9		X		
Fungus	3.9	4.7.6.10		X		
Life	3.5, 3.6.7.1	4.7.6.11	X	X	X	

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4.7 Test methods.

4.7.1 Operation. The encoder shall be connected in accordance with 4.3.1 when indicators (NAVORD Drawing 2639726) are used for monitoring. The rate of rotation for the test shall be greater than 80 counts per minute for the lowest decade being checked, but the rate of rotation shall not exceed 50 revolutions per minute. Should a different monitoring system be used, the rate of rotation shall not exceed the rate at which the monitoring system responds to each count of the decades being checked, or 50 revolutions per minute, whichever is less.

4.7.1.1 The shaft shall be rotated in the counterclockwise direction through a minimum of 36 revolutions and an increasing and correct count shall be verified. The units, tens, and hundreds decades shall be checked for consecutive counting for a minimum of one revolution. The hundreds, thousands, and ten-thousands decades shall be checked for consecutive counting for a minimum of 36 revolutions.

4.7.1.2 The procedure of 4.7.1.1 shall be repeated, except the shaft shall be rotated in the clockwise direction and shall cause a decreasing count.

4.7.2 Accuracy. The encoder shall be checked for accuracy by rotating the shaft in the counterclockwise direction on a precision divider head having a maximum error of 1 arc minute. Position readings shall be taken for each count change through one revolution of the shaft. Automated test methods may be used provided that the encoder drive speed is recorded, directly or indirectly, on the same recording medium as the encoder output. The resultant data shall be examined for the following:

(a) No more than one count change occurring in 10 arc minutes of rotation in one direction.

(b) The rotation from any count to any other count being 21.6n ± 21.6 arc minutes, where n is the numeric difference between the two counts.

4.7.3 Contact parameter measurement.

4.7.3.1 The encoder shall be connected as in 4.3.1. The voltage shall be measured between each encoder output pin and direct-current power ground on the decoder, with the encoder nonrotating and with contacts closed. The voltages across closed contacts shall also be measured at a shaft rotation speed of 50 revolutions per minute. Voltages shall be measured to an accuracy of ±0.1 volt.

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4.7.3.2 The encoder shall be disconnected from the configuration of 4.3.1. The resistance of each contact from negative polarity encoder output pin to the appropriate positive polarity common shall be measured when open, using a device capable of measuring at least 100,000 ohms with ±5 percent accuracy.

4.7.4 Insulation resistance measurement. The encoder shall be disconnected from the configuration of 4.3.1 and 500 ± 50 volts alternating current peak-to-peak or 200 ± 20 volts direct current applied between the case and each lead. The insulation resistance shall be measured using a device capable of measuring at least 10 megohms with ±5 percent accuracy as the input shaft is rotated through 72 revolutions at a rate of between 5 and 10 revolutions per minute.

4.7.5 Slew speed performance. The encoder shall be subjected to input shaft rotation rates of up to 200 revolutions per minute in each direction. After completion, the encoder shall be subjected to the procedures of 4.6 and 4.7.1.

4.7.6 Environmental tests.

4.7.6.1 High temperature, operating. The encoder, less packaging, shall be subjected to a temperature of 71.1° ± 1.7° Celsius for a period of 3 hours minimum. Within 15 minutes after removal from the temperature chamber, the encoder shall be subjected to the procedures of 4.7.1.1 and 4.7.1.2.

4.7.6.2 Low temperature, operating. The encoder, less packaging, shall be subjected to a temperature of -12.5° ± 2.5° Celsius for a period of 6 hours minimum. Within 15 minutes after removal from the temperature chamber, the encoder shall be subjected to the procedures of 4.7.1.1 and 4.7.1.2.

4.7.6.3 High temperature, storage. The encoder, less packaging, shall be subjected to a temperature of 82.5° ± 2.5° Celsius for a period of 24 hours minimum. After returning to room temperature, the encoder shall be subjected to the procedures of 4.6, 4.7.1, and 4.7.2.

4.7.6.4 Low temperature, storage. The encoder, less packaging, shall be subjected to a temperature of -58.5° ± 3.5° Celsius for a period of 72 hours minimum. After returning to room temperature, the encoder shall be subjected to the procedures of 4.6, 4.7.1, and 4.7.2.

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4.7.6.5 Vibration, operating. The encoder, less packaging, shall be subjected to the procedures of 4.7.1 and 4.7.2 during the test of type 1 mechanical vibration in MIL-STD-167.

4.7.6.6 Vibration, nonoperating. The encoder, less packaging, shall be subjected to vibration cycling in accordance with MIL-STD-810, method 514, procedure X, curve AY, for a period of 1 hour in each of the three axes. At the completion of this test, the encoder shall be subjected to the procedures of 4.6, 4.7.1, and 4.7.2.

4.7.6.7 Shock. The encoder, less packaging, shall be subjected to the shock test of MIL-STD-810, method 516, procedure I, figure 516.1-2, amplitude b, duration d. At the completion of this test, the encoder shall be subjected to the procedures of 4.6, 4.7.1, and 4.7.2.

4.7.6.8 Humidity. The encoder, less packaging, shall be subjected to the humidity tests of MIL-STD-810, method 507, procedure I, except that the the upper temperature shall be 52.7° ± 3.5° Celsius instead of 71° Celsius. Within 1 hour the encoder shall be subjected to the procedures of 4.7.1 and 4.7.2.

4.7.6.8.1 Humidity, operating. If specified by the procuring activity, the encoder shall be subjected to the procedures of 4.7.1 and 4.7.2, herein, while at 52.7° ± 3.5° Celsius and 85 percent relative humidity during the last cycle of the humidity test of 4.7.6.8.

4.7.6.9 Salt fog. The encoder shall be tested in accordance with MIL-STD-810, method 509, procedure I. After the test, the encoder shall be examined for evidence of corrosion. After drying and removal of salt deposits, the encoder shall be subjected to the procedures of 4.7.1 and 4.7.2.

4.7.6.10 Fungus. If specified by the procuring activity, the encoder shall be tested in accordance with MIL-STD-810, method 508, procedure I. After the test, the encoder shall be examined for evidence of deterioration, corrosion, or fungus growth, and the encoder shall be subjected to the procedures of 4.7.1 and 4.7.2.

4.7.6.11 Life. The encoder shaft shall be rotated clockwise for 15 minutes and counterclockwise for 15 minutes at 180 to 200 revolutions per minute. The alternate clockwise and counterclockwise rotation shall

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be continued until the shaft has been rotated 10,000,000 revolutions. After completion of each 500,000 revolutions, the encoder shall be subjected to the procedure of 4.7.1. The commutating power, as defined in 3.6.7.1, shall be measured to an accuracy of ±5 percent.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the procuring activity, the encoder shall be preserved in accordance with MIL-P-116, method II, and packaged and packed in accordance with MIL-STD-794.

5.2 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Shipment nomenclature for marking shall be as a minimum "ENCODER, NAVORD Drawing 2639621."

6. NOTES

6.1 Intended use. The encoder is intended for use in Test Set Mark 281 Mod 1. It provides coded shaft-angle position to the decoder which provides signals to the digital indicators for readout.

6.2 Ordering data. Procurement documents should specify, but not be restricted to, the following:

(a) Title, number, and date of this specification

(b) Encoder drawing number: NAVORD Drawing 2639621

(c) Quantity ordered should include test units that will be expended as a result of first article inspection

(d) Information concerning inclusion of optional environmental tests and/or exceptions to this specification or other pertinent documents

(e) Level of packaging and packing desired, if other than as specified in 5.1.

6.3 Output code description.

6.3.1 Cyclic decimal system. The cyclic decimal system is derived by changing the normal increasing only counting of each Arabic decimal decade

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to alternate increasing and decreasing counting of each decade. This system is shown in table II. Note that the units decade in the Arabic decimal always counts 0 to 9, but the units decade in the cyclic decimal system counts 0 to 9, 9 to 0, 0 to 9. This same increasing and decreasing counting sequence continues for all decades as the coded numbers get progressively larger.

Table II

EXAMPLE OF ARABIC DECIMAL TO CYCLIC DECIMAL CONVERSION

Arabic decimal	Cyclic decimal	Arabic decimal	Cyclic decimal	Arabic decimal	Cyclic decimal
00	00	10	19	20	20
01	01	11	18	21	21
02	02	12	17	22	22
03	03	13	16	23	23
04	04	14	15	24	24
05	05	15	14	25	25
06	06	16	13	26	26
07	07	17	12	27	27
08	08	18	11	28	28
09	09	19	10	29	29

6.3.1.1 Arabic to cyclic conversion. An Arabic decimal number is converted to a cyclic decimal number by the following rules:

(a) The most significant digit of any number is not changed.

(b) The second most significant digit is changed to a nine's complement if the most significant digit is odd; it is not changed if the most significant digit is even.

(c) The third most significant digit is changed to a nine's complement if the second most significant digit was odd before the conversion of (b); it is not changed if the second most significant digit was even before the conversion of (b).

(d) These rules are applied sequentially to all decades that a number has.

6.3.1.2 Cyclic to Arabic conversion. A cyclic decimal number is converted to an Arabic decimal number by the following rules:

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(a) The most significant digit of any number is not changed.

(b) The second most significant digit is changed to a nine's complement if the most significant digit is odd; it is not changed if the most significant digit is even.

(c) The third most significant digit is changed to a nine's complement if the second most significant digit is odd after the conversion of (b); it is not changed if the second most significant digit is even after the conversion of (b).

(d) These rules are applied sequentially to all decades that a number has.

6.3.2 Basic code. To eliminate the need for a separate brush for each digit of a decade, a four-bit code is used to generate the numbers of the cyclic decimal decades. The closed 1 state, and open 0 state, used in the code are defined in 3.6.9.1 and 3.6.9.2. The code structure for a full decade is given in table III.

Table III

CODED EQUIVALENTS FOR A CYCLIC DECIMAL DECADE

Cyclic decimal number	Coded equivalent			
	A Bit	B Bit	C Bit	D Bit
0	1	0	0	0
1	1	1	0	0
2	0	1	0	0
3	0	1	1	0
4	0	0	1	0
5	0	0	1	1
6	0	1	1	1
7	0	1	0	1
8	1	1	0	1
9	1	0	0	1

6.3.3 Disc coding rules. The basic cyclic decimal code is unambiguous provided that all decades are coded on one disc. However, the required encoder has two discs. Possible ambiguities of a two-disc system are eliminated by using a selector bit on the high-speed disc to control how the low-speed disc output is read.

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6.3.3.1 High-speed disc. The high-speed disc has the following coded bits and their complements: units A, B, C, D; tens A, B, C, D; hundreds A, B, C; and selector. All bit states are always interrogated between data output lines and encoder common. The correct 1 and 0 states of the selector bit shall conform to the following: 0 for the first half of the disc revolution, and 1 for the second half of the disc revolution. That is, the selector is 0 for 00000 to 00499, for 01000 to 01499, etc.; the selector is 1 for 00500 to 00999, for 01500 to 01999, etc. Additional examples of selector operation are given in table IV.

6.3.3.2 Low-speed disc. The low-speed disc has the following coded bits and their complements: hundreds D; thousands A, B, C, D; ten-thousands A, B, C. The correct 1 and 0 states of the A, B, C, D bits conform to the rules for coding cyclic decimal information. In arriving at the correct states, the following conditions are met:

(a) A brush cannot change state while it is under interrogation.

(b) When the cyclic code requires a change of state for a bit under interrogation, it is accomplished by electrically switching to a different brush that is in the desired state.

(c) Brush switching is accomplished by backward-biasing the series diode of the brush that is no longer to be interrogated and forward-biasing the series diode of the brush that is to be interrogated.

(d) Diode biasing is controlled by the decoder (NAVORD Drawing 2639792) in the following manner: The diodes in series with the brushes that are connected to the retard common (CR) will be forward-biased when the selector bit of the high-speed disc is a 1, and the diodes in series with the brushes that are connected to the advance common (CA) will be forward-biased when the selector bit of the high-speed disc is a 0.

(e) The correct states of bits are always capable of being interrogated between data lines and whichever common is forward-biasing its series diode.

6.3.4 Sample code transitions. An encoder coded in accordance with all the code requirements presented will be unambiguous; that is, there will be no more than one mechanical brush state change between any two consecutive numbers. To aid this examination, some examples of key number changes are given in table IV.

Custodian:
Navy - OS

Preparing activity:
Navy - OS
(Project Number 7440-N091)

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Table IV

CODED EQUIVALENTS

Arabic transition	Ten-thou with respect to CR	Ten-thou with respect to CA	Thou with respect to CR	Thou with respect to CA	Selector	Hundreds	Tens	Units
00099	0	Open	0	Open	0	0	9	0
00100	0	Open	0	Open	0	1	9	0
00199	0	Open	0	Open	0	1	0	0
00200	0	Open	0	Open	0	2	0	0
00499	0	Open	0	Open	0	4	9	0
00500	Open	0	Open	0	1	5	9	0
00599	Open	0	Open	0	1	5	0	0
00600	Open	0	Open	0	1	6	0	0
00999	Open	0	Open	0	1	9	0	0
01000	0	Open	1	Open	0	9	0	0
01499	0	Open	1	Open	0	5	9	0
01500	Open	0	Open	1	1	4	9	0
19999	Open	1	Open	0	1	0	0	0
20000	2	Open	0	Open	0	0	0	0
29999	Open	2	Open	9	1	0	0	0
30000	3	Open	9	Open	0	0	0	0
35999	Open	3	Open	4	1	0	0	0
00000	0	Open	0	Open	0	0	0	0

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-82578 (OS), Encoder, Shaft-Angle-Position, Cyclic-Decimal-Coded (for Torpedo Test Set Mark 281 Mod 1)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Attn: Standardization Division (QA3)
Indian Head, Maryland 20640

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